

Module 7: Aurora Assignment

Problem Statement: You work for XYZ Corporation. Their application requires a SQL service that can store data which can be retrieved if required. Implement a suitable RDS engine for the same. While migrating, you are asked to perform the following tasks:

1. Create an AuroraDB Engine based RDS Database.
2. Create 2 Read Replicas in different availability zones for better infrastructure availability

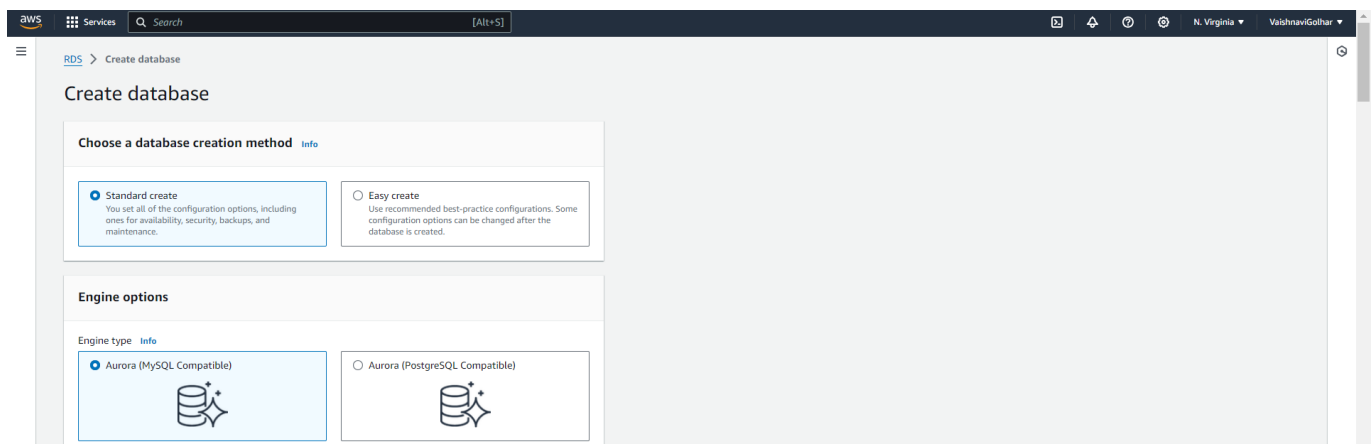
Solution:

Step 1: Create an AuroraDB Engine Based RDS Database

Log in to the AWS Management Console: Open the Amazon RDS console at <https://console.aws.amazon.com/rds/>.

Create Database: Click on “Create database”.

Engine Options: Select "Amazon Aurora" as the engine type.



Templates: Choose a template that suits your use case (Free tier, Dev/Test, or Production).

Edition: Select the edition of Aurora that meets your requirements (either Aurora MySQL or Aurora PostgreSQL).

DB Instance Class: Choose the appropriate instance class based on your workload and performance needs.

Multi-AZ Deployment: Enable Multi-AZ deployment for high availability. This ensures that your database has synchronous replication across multiple Availability Zones.

Storage: Configure your storage settings as needed.

Connectivity: Under Network & Security, select a VPC and configure the VPC security group(s) to allow inbound traffic on the Aurora port (usually 3306 for MySQL or 5432 for PostgreSQL).

Database Authentication: Choose your preferred authentication method (password authentication or IAM database authentication).

Additional Configuration: Provide a DB instance identifier, master username, and master password. You can also specify an initial database name if needed.

Create Database: Review all settings and click “Create database”. Wait for the database to be created.

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Instance configuration

The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class

Info

▼ Hide filters

☒ Include previous generation classes

☐ Serverless v2

☐ Memory optimized classes (includes r classes)

☒ Burstable classes (includes t classes)

db.t3.medium

2 vCPUs 4 GiB RAM Network: 2,085 Mbps

▼

Availability & durability

Multi-AZ deployment

Info

☒ Create an Aurora Replica or Reader node in a different AZ (recommended for scaled availability)
Creates an Aurora Replica for fast failover and high availability.

☐ Don't create an Aurora Replica

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Connectivity

info

Compute resource

Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☒ Don't connect to an EC2 compute resource
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☐ Connect to an EC2 compute resource
Set up a connection to an EC2 compute resource for this database.

Network type

info

To use dual-stack mode, make sure that you associate an IPv6 CIDR block with a subnet in the VPC you specify.

☒ IPv4
Your resources can communicate only over the IPv4 addressing protocol.

☐ Dual-stack mode
Your resources can communicate over IPv4, IPv6, or both.

Virtual private cloud (VPC)

info

Choose the VPC. The VPC defines the virtual networking environment for this DB cluster.

Default VPC (vpc-0b22f56ff603e7fa1)
6 Subnets, 6 Availability Zones

Only VPCs with a corresponding DB subnet group are listed.

After a database is created, you can't change its VPC.

DB subnet group

info

Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB cluster can use in the VPC that you selected.

default-vpc-0b22f56ff603e7fa1
6 Subnets, 6 Availability Zones

Public access

info

☒ Yes
RDS assigns a public IP address to the cluster. Amazon EC2 instances and other resources outside of the VPC can connect to your

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Yes

RDS assigns a public IP address to the cluster. Amazon EC2 instances and other resources outside of the VPC can connect to your cluster. Resources inside the VPC can also connect to the cluster. Choose one or more VPC security groups that specify which resources can connect to the cluster.

No

RDS doesn't assign a public IP address to the cluster. Only Amazon EC2 instances and other resources inside the VPC can connect to your cluster. Choose one or more VPC security groups that specify which resources can connect to the cluster.

VPC security group (firewall)

Info

Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

Choose existing

Choose existing VPC security groups

Create new

Create new VPC security group

Existing VPC security groups

Choose one or more options

default X

RDS Proxy

RDS Proxy is a fully managed, highly available database proxy that improves application scalability, resiliency, and security.

Create an RDS Proxy

Info

RDS automatically creates an IAM role and a Secrets Manager secret for the proxy. RDS Proxy has additional costs. For more information, see [Amazon RDS Proxy pricing](#).

Certificate authority - optional

Info

Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.

rds-ca-rsa2048-g1 (default)

Expiry: May 26, 2061

If you don't select a certificate authority, RDS chooses one for you.

Additional configuration

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If you don't select a certificate authority, RDS chooses one for you.

Additional configuration

Database port

Info

TCP/IP port that the database will use for application connections.

3306

Read replica write forwarding

Turn on local write forwarding

Info

Issues write operations from reader DB instances within the same DB cluster.

Database authentication

Info

Password authentication is always active for your database engine. You can also turn on additional authentication methods for your database below.

IAM database authentication

Authenticates using IAM database authentication.

Kerberos authentication

Authenticates using Kerberos authentication through an AWS Directory Service for Microsoft Active Directory.

Monitoring

Enable Enhanced Monitoring

Enabling Enhanced Monitoring metrics are useful when you want to see how different processes or threads use the CPU.

Amazon RDS console showing the "Creating database database-1" status. The console displays a notification about creating a Blue/Green Deployment to minimize downtime during upgrades. Below the notification, the "Databases (3)" section shows a table of database instances.

Consider creating a Blue/Green Deployment to minimize downtime during upgrades
You may want to consider using Amazon RDS Blue/Green Deployments and minimize your downtime during upgrades. A Blue/Green Deployment provides a staging environment for changes to production databases. [RDS User Guide](#) [Aurora User Guide](#)

Databases (3) Group resources Modify Actions Restore from S3 Create database

Filter by databases

DB identifier	Status	Role	Engine	Region & AZ	Size	Recommendations	CPU	Current activity	Mail
database-1	Creating	Regional cluster	Aurora MySQL	us-east-1	2 instances	-	-	non	
database-1-instance-1	Creating	Reader instance	Aurora MySQL	us-east-1d	db.t3.medium	-	-	non	
database-1-instance-1-us-east-1b	Creating	Reader instance	Aurora MySQL	us-east-1b	db.t3.medium	-	-	non	

XAMPP Control Panel v3.3.0 [Compiled: Apr 6th 2021]

XAMPP Control Panel v3.3.0

Modules

Service	Module	PID(s)	Port(s)	Actions
☐	Apache			Start Admin Config Logs
☒	MySQL	5368	3306	Stop Admin Config Logs
☐	FileZilla			Start Admin Config Logs
☐	Mercury			Start Admin Config Logs
☐	Tomcat			Start Admin Config Logs

Config Netstat Shell Explorer Services Help Quit

22:45:10 [main] XAMPP Installation Directory: "c:\xampp\
22:45:10 [main] Checking for prerequisites
22:45:31 [main] All prerequisites found
22:45:31 [main] Initializing Modules
22:45:31 [main] Starting Check-Timer
22:45:31 [main] Control Panel Ready
22:45:54 [mysql] Attempting to start MySQL app...
22:45:54 [mysql] Status change detected: running

Amazon RDS console showing details for database-1-instance-1.

Related

DB identifier	Status	Role	Engine	Region & AZ	Size	Recommendations	CPU	Current activity
database-1	Available	Regional cluster	Aurora MySQL	us-east-1	2 instances		-	-
database-1-instance-1	Available	Writer instance	Aurora MySQL	us-east-1d	db.t3.medium		40.85%	3 Select
database-1-instance-1-us-east-1b	Creating	Reader instance	Aurora MySQL	us-east-1b	db.t3.medium		-	-

Connectivity & security

Endpoint & port	Networking	Security
Endpoint: database-1-instance-1.ch0g2wak6pdx.us-east-1.rds.amazonaws.com Port: 3306	Availability Zone: us-east-1d VPC: vpc-0b22f56ff603e7fa1 Subnet group: default-vpc-0b22f56ff603e7fa1	VPC security groups: default (sg-08695198bc1703d9e) - Active Publicly accessible: Yes Certificate authority: rds-ca-rsa2048-g1

```
mysql -h database-1-instance-1.ch0g2wak6pdx.us-east-1.rds.amazonaws.com -u admin -p

Setting environment for using XAMPP for Windows.
Suttu@DESKTOP-A07FINS c:\xampp
# mysql -h database-1-instance-1.ch0g2wak6pdx.us-east-1.rds.amazonaws.com -u admin -p
Enter password: *****
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 69
Server version: 8.0.28 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

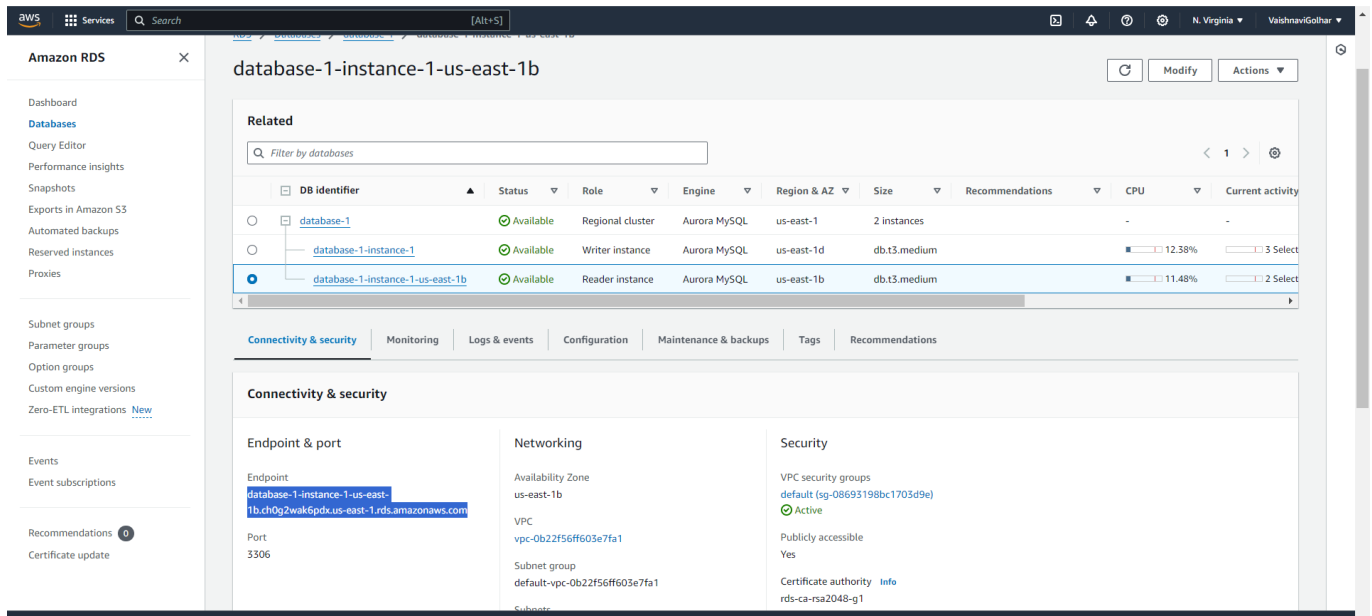
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> show databases;
+-----+
| Database |
+-----+
| Student |
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.227 sec)

MySQL [(none)]> use Student
Database changed
MySQL [Student]> create table class(name varchar(15));
Query OK, 0 rows affected (0.260 sec)

MySQL [Student]> show table;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '' at line 1
MySQL [Student]> show class
-> show databases;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'class
show databases;' at line 1
MySQL [Student]> show tables;
+-----+
| Tables_in_Student |
+-----+
| class |
+-----+
1 row in set (0.220 sec)

MySQL [Student]>
```



```
Suttu@DESKTOP-A07FIN5 c:\xampp
# mysql -h database-1-instance-1-us-east-1b.ch0g2wak6pdx-us-east-1.rds.amazonaws.com -u admin -p
Enter password: *****
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 57
Server version: 8.0.28 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> show databases;
+-----+
| Database |
+-----+
| Student |
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.232 sec)

MySQL [(none)]> use Student;
Database changed
MySQL [Student]> create table home(address varchar(50));
ERROR 1836 (HY000): Running in read-only mode
MySQL [Student]>
```

Step 2: Create 2 Read Replicas in Different Availability Zones

Navigate to the RDS Dashboard: Once your Aurora database is created, navigate to the RDS dashboard.

Select your Aurora database: Click on the name of your Aurora database to view its details.

Create Read Replica: Under the "Instance actions" dropdown menu, select "Create read replica".

Configure Read Replica:

Choose the appropriate instance class and availability zone for the first read replica.

Repeat the process to create a second read replica in a different availability zone.

Review and Create: Review all settings and click “Create read replica”. Wait for the read replicas to be created.

Amazon RDS console screenshot showing the details of a database instance named 'database-1'. The instance is in the 'Available' state. The 'Related' section displays a table with columns: DB identifier, Status, Role, Engine, Region & AZ, Size, and Recommendations. The table lists 'database-1' with status 'Available', role 'Regional cluster', engine 'Aurora MySQL', region 'us-east-1', and size '2 instances'. An 'Actions' menu is open, showing options like 'Stop temporarily', 'Delete', 'Set up EC2 connection', 'Set up Lambda connection', 'Add AWS Region', 'Add reader', and 'Create cross-Region read replica'.

Amazon RDS console screenshot showing the 'Add reader' configuration page. The page title is 'Add reader'. Below the title, it states: 'You are creating a replica DB instance from a source DB instance. This new DB instance will have the source DB instance's DB security groups and DB parameter groups.'

The configuration is divided into two main sections:

- Settings**
 - Aurora replica source**
 - Source DB cluster identifier: database-1-instance-1-us-east-1b
 - Role: Reader instance Parent: database-1
 - DB instance identifier**
 - This is the unique key that identifies a DB instance. This parameter is stored as a lowercase string (for example, mydbinstance).
 - Value: reader2
- Instance configuration**
 - The DB instance configuration options below are limited to those supported by the engine that you selected above.
 - DB instance class** [Info](#)
 - ▼ Hide filters
 - ☐ Include previous generation classes
 - ☐ Serverless v2
 - ☐ Memory optimized classes (includes r classes)
 - ☒ Burstable classes (includes t classes)

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db.t3.medium

2 vCPUs 4 GiB RAM Network: 2,085 Mbps

AWS Region

Destination Region

The Region where the replica will be launched.

US East (N. Virginia)

The replica database will be in the same Region as the source database.

Connectivity

Public access

☒ Publicly accessible

RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.

☐ Not publicly accessible

No IP address is assigned to the DB instance. EC2 instances and devices outside the VPC can't connect.

Availability Zone

Info

The EC2 Availability Zone that the database will be created in.

us-east-1c

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▼ Additional configuration

Database port

Info

Port number on which the database accepts connections

3306

Database authentication

Info

Password authentication is always active for your database engine. You can also turn on additional authentication methods for your database below.

☐ IAM database authentication

Authenticates using IAM database authentication.

☐ Kerberos authentication

Authenticates using Kerberos authentication through an AWS Directory Service for Microsoft Active Directory.

▼ Additional configuration

encryption turned off, failover, Enhanced Monitoring turned off, maintenance, CloudWatch Logs

DB parameter group

Info

default.aurora-mysql8.0

Failover priority

No preference

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Authenticate using Kerberos authentication through an AWS Directory Service for Microsoft Active Directory.

Additional configuration

encryption turned off, failover, Enhanced Monitoring turned off, maintenance, CloudWatch Logs

DB parameter group

Info

default.aurora-mysql8.0

Failover priority

No preference

Encryption

☐ Enable encryption

Choose to encrypt the given instance. Master key IDs and aliases appear in the list after they have been created using the AWS Key Management Service console. Info

☐ Enable Enhanced Monitoring

Enabling Enhanced Monitoring metrics are useful when you want to see how different processes or threads use the CPU.

Maintenance

Auto minor version upgrade Info

☐ Enable auto minor version upgrade

Enabling auto minor version upgrade will automatically upgrade to new minor versions as they are released. The automatic upgrades occur during the maintenance window for the database.

Cancel

Add reader

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Amazon RDS

Creating replica reader2 in US East (N. Virginia)

Your database might take a few minutes to launch.

Dashboard

Databases

Query Editor

Performance insights

Snapshots

Exports in Amazon S3

Automated backups

Reserved instances

Proxies

Subnet groups

Parameter groups

Option groups

Custom engine versions

Zero-ETL integrations New

Events

RDS > Databases

Consider creating a Blue/Green Deployment to minimize downtime during upgrades

You may want to consider using Amazon RDS Blue/Green Deployments and minimize your downtime during upgrades. A Blue/Green Deployment provides a staging environment for changes to production databases. RDS User Guide Aurora User Guide

Databases (4)

Group resources

Modify

Actions

Restore from S3

Create database

Filter by databases

< 1 >

DB identifier	Status	Role	Engine	Region & AZ	Size	Recommendations	CPU	Current activity
database-1	Available	Regional cluster	Aurora MySQL	us-east-1	3 instances		-	-
database-1-instance-1	Available	Writer instance	Aurora MySQL	us-east-1d	db.t3.medium		12.73%	3 Selects/S
database-1-instance-1-us-east-1b	Available	Reader instance	Aurora MySQL	us-east-1b	db.t3.medium		12.67%	2 Selects/S
reader2	Creating	Reader instance	Aurora MySQL	us-east-1c	db.t3.medium		-	-